

===== Start of q1.c =====

```
#include <stdio.h>
```

```
int main() {
    int a[5], i, j, temp;

    for(i = 0; i < 5; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < 4; i++)
        for(j = i + 1; j < 5; j++)
            if(a[i] > a[j]) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }

    for(i = 0; i < 5; i++)
        printf("%d ", a[i]);

    return 0;
}
```

===== End of q1.c =====

===== Start of q2.c =====

```
#include <stdio.h>
```

```
int main() {
    int a[5], i, j, temp;

    for(i = 0; i < 5; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < 4; i++) {
        for(j = i + 1; j < 5; j++) {
            if(a[i] > a[j]) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }

    printf("Second largest element: %d\n", a[3]);
}
```

```
    return 0;
}
```

===== End of q2.c =====

===== Start of q3.c =====

```
#include <stdio.h>
```

```
int main() {
    int a[3][4], i, j;

    for(i = 0; i < 3; i++)
        for(j = 0; j < 4; j++)
            scanf("%d", &a[i][j]);

    for(i = 0; i < 3; i++) {
        for(j = 0; j < 4; j++)
            printf("%d\t", a[i][j]);
        printf("\n");
    }
    return 0;
}
```

===== End of q3.c =====

===== Start of q4.c =====

```
#include <stdio.h>
```

```
int main() {
    int a[3][4], i, j, sum = 0;

    for(i = 0; i < 3; i++)
        for(j = 0; j < 4; j++) {
            scanf("%d", &a[i][j]);
            sum += a[i][j];
        }

    printf("%d", sum);
    return 0;
}
```

===== End of q4.c =====

===== Start of q5.c =====

```
#include <stdio.h>
```

```
int main() {  
    int a[3][4], i, j;  
  
    for(i = 0; i < 3; i++)  
        for(j = 0; j < 4; j++)  
            scanf("%d", &a[i][j]);  
  
    for(i = 0; i < 3; i++)  
        for(j = 0; j < 4; j++)  
            if(a[i][j] % 3 == 0 || a[i][j] % 5 == 0)  
                printf("%d ", a[i][j]);  
  
    return 0;  
}
```

```
===== End of q5.c =====
```

```
===== Start of q6.c =====
```

```
#include <stdio.h>
```

```
int main() {  
    int a[3][4], i, j, k, flag;  
  
    for(i = 0; i < 3; i++)  
        for(j = 0; j < 4; j++)  
            scanf("%d", &a[i][j]);  
  
    for(i = 0; i < 3; i++)  
        for(j = 0; j < 4; j++) {  
            if(a[i][j] < 2) continue;  
            flag = 1;  
            for(k = 2; k <= a[i][j] / 2; k++) {  
                if(a[i][j] % k == 0) {  
                    flag = 0;  
                    break;  
                }  
            }  
            if(flag)  
                printf("%d ", a[i][j]);  
        }  
  
    return 0;  
}
```

===== End of q6.c =====

===== Start of q7.c =====

```
#include <stdio.h>
int main(){
    int a[3][4], b[4][4];
    printf("enter the array elements:\n");
    for (int i=0; i<3; i++){
        for (int j=0; j<4; j++){
            printf("A[%d][%d]: ", i+1, j+1);
            scanf("%d",&a[i][j]);
        }
    }
    printf("Original Matrix:\n");
    for (int i=0; i<3; i++){
        for (int j=0; j<4; j++){
            printf("%d\t", a[i][j]);
            b[j][i]=a[i][j];
        }
        printf("\n");
    }
    printf("Transpose of the original Matrix:\n");
    for (int i=0; i<4; i++){
        for (int j=0; j<3; j++){
            printf("%d\t", b[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

===== End of q7.c =====

===== Start of q8.c =====

```
#include <stdio.h>

int main() {
    int a[3][4], sum;

    printf("Array elements:\n");
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 4; j++) {
            printf("a[%d][%d]: ", i + 1, j + 1);
            scanf("%d", &a[i][j]);
        }
    }
}
```

```

    }
}

printf("\nSum of rows:\n");
for (int i = 0; i < 3; i++) {
    sum = 0; // Reset sum for each new row
    for (int j = 0; j < 4; j++) {
        printf("%d\t", a[i][j]);
        sum += a[i][j];
    }
    printf("= %d\n", sum);
}

return 0;
}

```

===== End of q8.c =====

===== Start of q9.c =====
#include <stdio.h>

```

int main() {
    int a[3][4], b[4][3], sum;

    printf("Enter array elements:\n");
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 4; j++) {
            printf("a[%d][%d] ", i + 1, j + 1);
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nTransposing for even display...\n");
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 4; j++) {
            b[j][i] = a[i][j];
        }
    }

    printf("\nSum of column of a[3][4] (rows of b[4][3]):\n");
    for (int i = 0; i < 4; i++) {
        sum = 0;
        for (int j = 0; j < 3; j++) {
            printf("%d\t", b[i][j]);
            sum += b[i][j];
        }
    }
}

```

```

    }
    printf("(= %d)\n", sum);
}

return 0;
}

```

===== End of q9.c =====

===== Start of q10.c =====

```
#include <stdio.h>
```

```

int main() {
    int a[3][3], b[3][3], add[3][3], sub[3][3], mul[3][3];
    int i, j, k;

```

```

    // Input matrix A
    printf("Enter matrix A (3x3):\n");
    for(i = 0; i < 3; i++)
        for(j = 0; j < 3; j++)
            scanf("%d", &a[i][j]);

```

```

    // Input matrix B
    printf("Enter matrix B (3x3):\n");
    for(i = 0; i < 3; i++)
        for(j = 0; j < 3; j++)
            scanf("%d", &b[i][j]);

```

```

    // Perform operations
    for(i = 0; i < 3; i++)
        for(j = 0; j < 3; j++) {
            add[i][j] = a[i][j] + b[i][j];
            sub[i][j] = a[i][j] - b[i][j];
            mul[i][j] = 0;
            for(k = 0; k < 3; k++)
                mul[i][j] += a[i][k] * b[k][j];
        }

```

```

    // Print matrix A
    printf("\nMatrix A:\n");
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }

```

```

// Print matrix B
printf("\nMatrix B:\n");
for(i = 0; i < 3; i++) {
    for(j = 0; j < 3; j++)
        printf("%d ", b[i][j]);
    printf("\n");
}

// Print addition
printf("\nA + B:\n");
for(i = 0; i < 3; i++) {
    for(j = 0; j < 3; j++)
        printf("%d ", add[i][j]);
    printf("\n");
}

// Print subtraction
printf("\nA - B:\n");
for(i = 0; i < 3; i++) {
    for(j = 0; j < 3; j++)
        printf("%d ", sub[i][j]);
    printf("\n");
}

// Print multiplication
printf("\nA * B:\n");
for(i = 0; i < 3; i++) {
    for(j = 0; j < 3; j++)
        printf("%d ", mul[i][j]);
    printf("\n");
}

return 0;
}

```

===== End of q10.c =====

===== Start of q11.c =====

```
#include <stdio.h>
```

```
int main() {
    int a[4][3], i, j;
```

```
    // Input matrix
```

```

printf("Enter 4x3 matrix:\n");
for(i = 0; i < 4; i++)
    for(j = 0; j < 3; j++)
        scanf("%d", &a[i][j]);

// Display modified matrix
printf("\nModified matrix (odd as-is, even as -):\n");
for(i = 0; i < 4; i++) {
    for(j = 0; j < 3; j++) {
        if(a[i][j] % 2 != 0)
            printf("%d ", a[i][j]); // odd stays
        else
            printf("- ");           // even replaced by -
    }
    printf("\n");
}

return 0;
}

```

===== End of q11.c =====

===== Start of q12.c =====

```

#include <stdio.h>

int main(){
    int a[4][4], sum=0;
    printf("Enter array elements:\n");
    for(int i=0; i<4;i++){
        for (int j=0; j<4; j++){
            printf("A[%d][%d]: ", i+1, j+1);
            scanf("%d", &a[i][j]);
        }
    }
    for (int i=0; i<4; i++){
        sum+=a[i][i];
    }

    printf("\nSum of Diagonal elements: %d", sum);
    return 0;
}

===== End of q12.c =====

```